

OpenMath • Chair review book

3 October 2026 | Compact proposals and scientific frontier

Start here. Review each available packet, approve or hold its precise contribution, then reconcile the problem profiles. This book covers all70 grouped records /74 source identities and226 contribution rows. They are not70 verified registered teams. Shared authors receive one entrant score; personal point splits are not invented.

Review plan: 207 minutes for entries,60 for problem frontiers,45 for decisions and45 for breaks:357 minutes. Sakana and HTPeo need about20 minutes each; other material-bearing entries about10; missing/withdrawn accounts about1. Deep independent proof rebuilding and literature clearance cannot all be completed in that time.

Proposed tally — material-bearing entries

Entry / authors	S	A	M2 / adjusted	Disposition
E02 Rachel Teo / Wenyi Wang / Hamdi Abderrahmene	1697.401	308.025	10 / 10	Late ZIP / priority / new D
E03 Matt Bowring	0.000	0.000	0 / 0	Unproved universal bound
E04 Chandragupt Sharma — matrix support 138	28.830	28.830	0 / 0	Tensor verified; closure/priority
E05 Chaewon Yoon	6.556	6.556	0 / 0	Scoped novelty / closure checks
E06 Leanification — five-person IMPA Tech team	0.000	0.000	1 / 0.8	Five-person exception ×0.8
E08 HTPeo — Woohyuk Kang / Hyunjin Lee / Sanghyeon Lee / Youchan Oh	88.275	49.421	10 / 10	Scoped closures / novel classes
E09 Raj Harshit Srirangam	0.000	0.000	0 / 0	No formal geometry proof; potential17.870
E10 Jamie Steeg / grandchallenge / GCL	6.556	6.556	1 / 1	Scoped novelty / closure checks
E11 Leon Koerbs / Cameron Fen — Erdős #3 Token Burner	0.000	0.000	3 / 3	Scoped novelty / closure checks
E12 Frederik Smits van Oyen — K4 Ramsey and Heilbronn	0.000	0.000	0 / 0	Global cover incomplete
E17 Qichao Wang — Černý partial formalizations	0.000	0.000	0 / 0	Source missing; potential5 M2
E57 Rohith Poola — Kobon only	3.574	3.574	0 / 0	Scoped novelty / closure checks
E65 Wilson Wu — Erdős supporting formalizations	0.000	0.000	2 / 2	Scoped novelty / closure checks

S and A are original-result review proposals. M2 counts are separate, contingent on faithful reusable novelty. No awards or official acceptance have been recorded. All other roster records currently have0 original points and0 M2.

How to approve these proposals

One family: $F = b \times m \times p \times D^2 / 1000$. S is the sum of distinct family unions; A is the largest. Frozen focus $b=1.1$; other original targets $b=1$; new substantive variations $m=0.5$. Ramsey uses conservative $b=1$ here until its frozen focus inclusion is verified; confirmed focus would change both Ramsey families from 35.280 to 38.808.

Progress starts at the lower band endpoint: $P_0=0$; $P_1=.01-.05$; $P_2=.05-.15$; $P_3=.15-.35$; $P_4=.35-.60$; $P_5=.60-.85$; $P_6=.85-.95$; complete exact target=1. No value strictly between .95 and 1. A complete small subquestion can be $p=1$ with its own small D ; a narrowed restatement does not turn a partial parent solution into a full solution.

OPDP extends to every attempted problem. Nine existing published competition-scale profiles are retained; 61 local profiles use the same CFSD1000 methodology. Intrinsic /10 is one input, not $D/100$. Every local profile records its canonical scope, parent, source, inputs, calculation and axis rationale. New D values are one-assessor proposals: the required three-assessor median and two-judge progress decision have not been fabricated.

Formalization is assessed by reachable endpoint, statement fidelity, axioms and exact certificate coverage. Static source reading and supplied PASS receipts are identified as such. No contestant code was executed and no fresh Lean build is claimed. Unproved hypotheses, incomplete geometry transfer, missing vectors and uncovered boxes remain explicit holds.

Approval decisions with the largest effect

Decision	Proposed treatment
Sakana delivery	PDF received 05:57:20 CEST, before 06:00 cutoff; ZIP 06:19:17, after cutoff. Decide whether immutable already-finished sources and administrative transfer are acceptable; internal build dates are not independent submission receipts.
Half-turn and +3 hypergraph targets	Propose M3A: the exact questions predate the event. If classified M3B instead, subtract 54.450 and 154.0125 from Sakana respectively. Short proofs still require priority checks.
New D / focus list	Ratify source binding and canonical scope; get required independent D assessments. Ramsey bonus excluded until frozen-focus evidence is confirmed.
Known-math M2	Propose Sakana10, HTPeo10, Leon/Cameron3, Wilson2, Jamie1, Leanification1. HTPeo7 minor candidates held; 6 wrappers excluded. Qichao0 current, 5 potential groups with source missing.
Leanification	All 5 contributors are permitted by Alejandro. Raw M2 count 1; adjusted overall ranking value 0.8. Keep raw and adjusted values separate; no organizer acknowledgement invented.
Missing/incomplete mathematical payloads	Chandra original-source hold resolved: full 1047 file snapshot and exact tensors/family checked, proposed 28.82968 pending fresh closure/priority. Raj0 pending formal geometry transfer; Matt0 unproved universal synthesis; Heilbronn0 incomplete cover.
Independence / collective frontier	Sealed independent submissions may each earn entrant credit; the same scientific result is counted once. Similarity, equal score or public baseline exposure does not by itself prove copying.

Record decisions in score-signoff.csv and contribution-signoff.csv, or the local app's saved Review notes. Full contribution notes: review/entries/<resultID>.txt. Full OPDP and problem analyses: review/problems/<family>.txt.

E02 • Rachel Teo / Wenyi Wang / Hamdi Abderrahmene

Shared authors: Rachel Teo, Wenyi Wang, Hamdi Abderrahmene

Proposed S 1697.401 | A 308.025 | M2 10 / adjusted 10 | 20 minutes

Broadest original-result portfolio. Prioritize the two OEIS complete targets, then sharp pre-existing subquestions and uniform bounds. Its22Claim cards and5Focus cards are27dossiers, not27open solutions. Separate10known-math formalizations, four completed narrower targets and unfinished parent problems. On-time PDF05:57 and ZIP06:19 require a documented fixed-before-cutoff transfer decision.

Pinned commit: None | Repo: SakanaAI/autolab-100

Contribution / precise assessment	Novelty / formalization / proposed score
E02-a100475 • A100475: no positive prime-reversal cycle The positive-start restriction is essential. Growth of the prime-index map plus finite exceptional checks can rule out periodic tails for every positive start. Zero-start totalization would answer a different question. This is a complete target candidate, with a substantial source chain for a deceptively short statement.	Novelty: Candidate original resolution; bounded searches found no exact positive-start solution. Ball–Coxeter and the 2004 SeqFan discussion remain specific priority... Formal: Packet evidence: Independent complete 84-module replay completed 2026-10-01T13:03:31.027744Z; source receipt records byte identity with user ZIP; final theorem standard axioms.... M3A · D 541 · p 1 · 292.681 points before family union
E02-a060957 • A060957: refutation and arbitrarily long prime-exponent gaps This refutes the actual whole-interval distinct-subset-product interpolation assertion, not an arbitrary restricted factor set. The arbitrary-gap theorem is materially stronger but stays in this family. Prime-coordinate encoding and the exact eight-level fibre are the useful reusable mechanism.	Novelty: Strong original-result candidate against the 2020 OEIS conjecture; the located prior work supplies methods, not this complete-interval counterexample.... Formal: Packet evidence: Combined 45-module 4.33.1 replay and aggregate passed 2026-10-01 around16:08UTC. Generalized24-module closure also passed4.34.1 around16:01 and4.33.1 around16:05.... M3A · D 527 · p 1 · 277.729 points before family union
E02-a000224 • A000224: verified squarefree three-prime exclusion The uniform product-of-three-distinct-odd-primes theorem removes an infinite subcase. It does not handle repeated prime powers, four or more factors, or the entire OEIS equivalence. The exact definition counts zero as a square residue.	Novelty: Plausible new infinite-family advance; the located prior Pell scan is bounded and does not imply this theorem. Formal: Packet evidence: All 171 intended modules were recorded checked on 4.34.1, dependency-ordered and incremental; nine prime modules were reused after source/output hash checks. A... M3A · D 622 · p 0.15 · 58.033 points before family union
E02-ferrers-hierarchy • Ferrers general deficiency hierarchy (44-module dual replay) Preservation of an arbitrary supplied optimal factor is stronger than mere existence of one convenient factor. The hierarchy determines later matching-union ranks, while the $e > 1$ case does not realize every individual early CDS colour. That last limitation prevents a full Latin-tableau conclusion.	Novelty: Candidate new structural extension beyond known boundary cases. Some factor-theory full texts remain an explicit literature gap. Formal: Packet evidence: 44-source strict replay receipts on both pins; source hashes match both recorded receipts. Actual YoungDiagram graph/rank transport and all selected final... M3A · D 650 · p 0.15 · 63.375 points before family union
E02-erdos169-fourap • Explicit four-AP-free reciprocal sum $> 111/25$ The same finite set is used in positivity, four-AP freeness and the rational harmonic sum. A sum above4.44 is stronger than Walker's4.43975 and the located later4.4397534742 claim. It supplies a better lower bound, not unbounded sums or a solution of Erdős3.	Novelty: Candidate new numerical bound with exact formal witness; digit-restricted and block-combination methods are prior mathematics. Formal: Packet evidence: Fresh Linux full build completed8950jobs and final axiom audit. Exact same finite set appears in positivity, AP-free and rational-sum clauses. Mathematical source... M3A · D 538 · p 0.05 · 14.472 points before family union
E02-opdp13-quartic • Quartic exact exponent classification and sharp boundary scale The precise quartic family has both an exact exponent-semigroup classification and a matching asymptotic threshold $\delta \cdot N / \log(1/\delta)$ tending to one. The source uses a function-level holomorphic squared-norm representation, which is stronger than a coefficient-only claim.	Novelty: Original variation candidate. Tan–To's inspected appendix concerns multiplication by R^m , a different operation, but its main theorem and Handelman full text... Formal: Packet evidence: Final standalone40-proof-module and aggregate build completed3553jobs with exit0; source SHA records match; endpoint axioms standard. Earlier dependency-graph... M3B · D 644 · p 1 · 207.368 points before family union

Contribution / precise assessment	Novelty / formalization / proposed score
E02-opdp43-halfturn • Sharp half-turn symmetry defect gap A $2n$-point no-three-collinear grid set has exactly two points per row and column. The incidence discrepancy with its half-turn image has zero row and column balances, so a nonempty discrepancy must include an alternating cycle and at least four one-sided misses. The ten-point 5×5 witness attains four; symmetric difference is eight.	Novelty: The near-symmetry question is present on the pre-event Bielefeld source. The argument is short and has material folklore risk, but that is not proof of prior... Formal: Packet evidence: Clean4-module standalone release build completed 3097 jobs; source SHA entries match; final universal theorem includes $n_{0,1,2}$ and exact witness geometry; standard... M3A · D 330 · p 1 · 108.900 points before family union
E02-erdos1060-uniform-partial • Erdős 1060 uniform quantitative partial bound The proposed uniform exponent constant is about .363272, below $\log(3)/3$ about .366204. The separately located odd-target constant $\log(2)/2$ about .346574 applies only to odd n and does not subsume this all-target theorem. A positive constant still misses the requested little-o exponent.	Novelty: Candidate new all-target refinement; isolate the all-n domain from prior odd-only bounds and known powerful-core machinery. Formal: Packet evidence: Fresh isolated Lean 4.33.1 compilation passed all 10 own source modules, with no own olean reuse. Before/after source hashes agree and final canonical theorem... M3A · D 574 · p 0.05 · 16.474 points before family union
E02-erdos21-sharp-residual • Erdős Lovász sharp residual cover bound The $+3$ residual question is explicitly asked in the July 2026 paper, so it predates the competition. Closing it changes the elementary all-r bound from $3r-4$ to $3r-3$. The source's stronger approximately $3.053r$ asymptotic bound is not improved. Imported matching infrastructure must remain attributed.	Novelty: Strong explicit-subquestion candidate: the inspected primary paper asks for $+3$ after its $+4$ lemma. The known triangle sharpness example and imported author... Formal: Packet evidence: An independent isolated Lean 4.34.1 replay passed all 18 Main-closure modules. The proof-output directory began empty, source hashes stayed unchanged, nine... M3A · D 555 · p 1 · 308.025 points before family union
E02-erdos944-order-bound • Erdős 944 finite order obstruction The linear constraint $n \geq (27r+19)/4$ beats $5r+6$ for integer $r \geq 1$ after rounding, tying at $r=1$. The source's $n/(\log n)^c$ asymptotic restriction remains stronger eventually. This is a useful explicit finite-regime necessary condition, not a new construction or full existence answer.	Novelty: Candidate finite-regime improvement over the explicit prior bound. Existing $r=1$ graph constructions and classical coding inequalities receive no... Formal: Packet evidence: Independent isolated Lean 4.34.1 replay compiled three proof modules and the separate TargetAudit.lean from unchanged frozen sources. Its own output directory... M3A · D 721 · p 0.05 · 25.992 points before family union
E02-opdp89-parity-local-maxima • OPDP89 sharp four parity likelihood classification The theorem concerns the full nine-dimensional four-visible/one-hidden RBM, not only a one-dimensional ray. The strict threshold region has non-strict local maxima; equality and beyond are saddles. Every collapsed point is still suboptimal globally. The degenerate higher-order analysis is the substantive contribution.	Novelty: Candidate exact new threshold; collapsed stationarity, singular mixture lines and interaction formulas are prior work. No claim for general training landscapes... Formal: Packet evidence: Fresh post-recovery source-only replay completed 2026-10-02T18:45:45.269248+00:00 in 108.552 seconds. All 17 frozen source modules passed with no warnings or... M3B · D 727 · p 1 · 264.264 points before family union
E02-erdos829-log-five-thirds • Erdős 829 verified $\log(5/3)$ partial bound The all-target bound uses coefficient $\log(5/3)$, about .510826, rather than the earlier same-family $\log(7/4)$, about .559616. The located .556477 comparison is only cubefree and does not cover all targets. Including zero coordinates and ordered pairs preserves the original count. The existential threshold is not effective.	Novelty: Candidate new uniform bound built from prior ratio-root, determinant and coding methods. Earlier intermediate coefficients are revisions, not additional... Formal: Packet evidence: Fresh separately orchestrated source-only Lean 4.33.1 replay completed 2026-10-02T18:54:57.202016+00:00. All 33 proof modules and two semantic/axiom audit modules... M3A · D 530 · p 0.05 · 14.045 points before family union
E02-FOCUS-E3 • Erdős 3: finite diagonal-pullback nonoptimality The modulus 13 witness 79 beats the best diagonal-pullback capacity 78. It disproves that proposed finite auxiliary extremal ansatz. It supplies no divergent infinite set and no theorem forcing longer arithmetic progressions.	Novelty: Candidate finite obstruction to a particular ansatz; direct original-problem pivotality is limited. Formal: The cyclic sharpness certificate and source-faithful diagonal-pullback bridge passed Lean 4.33.1 / Mathlib 0df444a360eaa60ab8c11dca51a86af692955474 with standard axioms only. The... M1 · D 674 · p 0.01 · 4.997 points before family union

Contribution / precise assessment	Novelty / formalization / proposed score
E02-FOCUS-GROTH • Grothendieck: restricted sign-matrix bounds Three-row sign matrices and four archived finite matrices cannot control all real-coefficient Grothendieck instances. The exact finite inequalities are reusable certificates, but no new bound on the universal constant follows.	Novelty: No original universal-constant advance established. A new formalization may be M2 only after source and prior-library comparison. Formal: Fresh Row3Granular and four-case SemanticTestGranular checks passed in Lean 4.33.1 with standard axioms only. Row3 mathematical body is byte-identical after import adjustment; the... M1 · D 561 · p 0 · 0.000 points before family union
E02-FOCUS-MATRIX • Matrix multiplication: one-leg rigidity The rank23 paired-factor matrices give uniqueness of the remaining factor at three known support139 seeds. This is a precise obstruction to changing only one factor list. It does not prohibit simultaneous changes of all lists, so it is compatible with Chandra's138-support claim.	Novelty: Candidate new exact method obstruction; the seeds are known and the linear-algebra route is classical. Formal: All three stored HKS139 seeds have standard-kernel certificates for the UV, UW and VW rational injectivity and remaining-factor uniqueness statements. Seed 0 retains its recorded... M1 · D 724 · p 0.01 · 5.766 points before family union
E02-FOCUS-RAMSEY • Ramsey: exact weighted construction and formal lifting The weighted bound is below the McKay reference0.0301422734319. HTPeo's reported rational is smaller again. Each has a reusable finite-to-limit lifting argument; independent sealed provenance, rather than merely uploading after the other, controls duplicate entrant credit.	Novelty: Exact bound improvement over the located2024 final-note reference. A bounded search found no stronger prior source; a separate Technion lead remains unresolved. Formal: The numerator is independently checked by exact C++/Python arithmetic and matches a nonfinal validation receipt. The general count-to-limit development freshly compiled after four... M3A · D 840 · p 0.05 · 35.280 points before family union
E02-FOCUS-BB6 • Busy Beaver 6: exact halting witness The six-state witness halts after1,235,211 steps with1,519 ones. That is a finite exact run, not maximality among all six-state machines. Head and visited span were separate simulator measurements, not the claimed kernel endpoint.	Novelty: No new original Busy Beaver frontier demonstrated; a reusable newly formalized simulator would require an explicit M2 claim and prior-library check. Formal: A new standard-kernel certificate in Lean 4.33.1 successfully checks the literal machine, exact run, final ones count, absence of an earlier halt and all-six-states visitation.... M1 · D 974 · p 0 · 0.000 points before family union

Known-mathematics formalizations

One family per admitted source theorem or reusable formal target; supporting lemmas remain in its union.

Result ID / target	Scope / review focus
E02-a046969-i • A046969 Conjecture I: exact Bernoulli denominator	For $a(n)=\text{den}(B_{(2n)/(2n-1)})$ when $n>0$ ($a(0)=0$), if $p>3$ and both p and $2p-1$ are prime, then $a(p)=12(2p-1)$; consequently $a(p)/12$ is prime.
E02-opdp67-quaternion • OPDP67 explicit two-witness quaternion scalar definition	For every linearly ordered commutative ring R satisfying <code>CommRing</code> , <code>LinearOrder</code> and <code>IsStrictOrderedRing</code> , a Hamilton quaternion q over R is scalar iff there exist unrestricted quaternions u,v with $((uv-vu)^2+4)^2-(qu-uv)^2-(qv-vq)^2=0$
E02-a135508 • A135508: prime-index gcd criterion and residue classes	Set $x(0)=0$, $x(1)=1$ and $x(n+1)=2x(n)+\text{lcm}(x(n),n+1)$ for $n\geq 1$; for $n>0$ let $a(n)=x(n+1)/x(n)-2$. For each prime $p>3$, $a(p-1)=p$ iff $\gcd(x(p-3),p-2)>1$. If $p\geq 7$ is prime and $p \bmod 3=2$ or $p \bmod 5=2$, then $a(p-1)=p$. More generally an acquired...
E02-apery-odd-formalization • A005258 Apéry prime-power irreducibility	For $B_n(X)=\sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k} X^k$ in $\mathbb{Q}[X]$, every odd prime p and $r\geq 1$ satisfying $p^2 \nmid 2^{p-1}-1$ has $\text{Irreducible}(B_{(p^r)})$. This card requests review only of any genuinely new useful formalization beyond the prior exposed...
E02-m2-research-base3-reverse-add • Base three reverse and add carry obstruction	If x plus its canonical base-3 digit reversal has palindromic base-3 digits, then x already has palindromic base-3 digits or every pair of mirrored input digits sums to less than 3.
E02-m2-research-theta-half-sum • Theta half sum negative inverse cancellation	For an odd prime k and odd natural a,b with $1\leq a,b < k$ and $ab \bmod k = k-1$, define $H(k,a)=\sum_{j=1}^{(k-1)/2} (-1)^{\text{floor}(aj/k)}$. Then $H(k,a)+H(k,b)=0$.

Result ID / target	Scope / review focus
E02-m2-research-wolstenholme • Reciprocal square numerator arithmetic	For every $m \geq 1$, the rational sum Σ from $k=1$ to m of $1/k^2$ has nonpositive 2-adic and 3-adic valuations. The selected bridge excludes a prime from its numerator when the valuation is nonpositive; for every prime $p > 3, \dots$
E02-m2-research-a060841 • A060841 reciprocal determinant integrality	For $n \geq 1$, the inverse determinant of the n by n matrix with entries $1/\text{lcm}(i+1, j+1)$ is an integer exactly when n belongs to $\{1, \dots, 34, 36, 38\}$.
E02-m2-research-erdos887 • Erdős and Rosenfeld four nearby divisors	There are infinitely many natural n with at least four distinct divisors d in the open real interval $(\sqrt{n}, \sqrt{n} + 40 n^{1/4})$. The preserved statement uses equivalent floor/ceiling integer endpoints.
E02-m2-research-regular-polygon • Regular polygon chord and distance multiplicity	For every even $n \geq 5$ and every distance d occurring between distinct vertices of the regular n -gon, the number of unordered vertex pairs at distance d is at most n .

Sign-off focus: Late ZIP 06:19:17 versus on-time PDF 05:57:20: prove fixed cutoff artifact or organizer admin-transfer decision; Exact source/library closure fresh independent build and semantic review; Target admission/open-at-freeze/novelty and canonicalID mapping; Ball–Coxeter 13th edition, pp.14–15; November and December 2004 SeqFan archive messages

E03 • Matt Bowring

Shared authors: Matt Bowring

Proposed S 0.000 | A 0.000 | M2 0 / adjusted 0 | 10 minutes

Interesting synthesis coefficient, but presently fails the formal-only rule. Review the exact claimed scope and move on; numerical circuits do not justify a provisional mathematical award.

Pinned commit: 57d6c96992a58f43d48d61c2a44353d902813cda | Repo: bowrango/lean-unitary

Contribution / precise assessment	Novelty / formalization / proposed score
E03-SYNTHESIS • Improved synthesis bound The21/48 formula would improve the2024 constructive22/48 bound, but numerical18-CNOT three-qubit fits do not establish coverage of every unitary. A generic Jacobian rank check gives at most local coverage, and parameters equal to the target dimension do not prove global surjectivity. The contestant acknowledges incomplete rigorous proof and no Lean.	Novelty: Potential new coefficient if proved. Compare to Krol–Al-Ars2024, not only the older23/48 baseline. Formal: Author acknowledges no Lean and incomplete rigorous proof; numerical circuits and parameter counting cannot establish generic all-n coverage. M3A · D 824 · p 0 · 0.000 points before family union

Sign-off focus: Rigorous constructive proof and approved formalization; Generic surjectivity/exceptional parameter treatment; Cutoff version: 57d6c969, later 68122d1 is not automatically eligible

E04 • Chandragupt Sharma — matrix support 138

Shared authors: Chandragupt Sharma

Proposed S 28.830 | A 28.830 | M2 0 / adjusted 0 | 10 minutes

Complete pinned original now retrieved:1047files. All729 identities independently verified for138,gauged138 and143, with exact Lean-data correspondence; symbolic rational family checked too. Proposed28.82968 for one improved-support family. No minimum-rank, global137 exclusion or runtime improvement established. Fresh closure, prior/independence and official intake remain.

Pinned commit: 479c2416b6261ee841052eef4881df0e40054f3f | Repo: ChandraguptSharma07/matrix-multiplication-tensor-3x3

Contribution / precise assessment	Novelty / formalization / proposed score
E04-SUPPORT138 • Rank-23 support-138 scheme The complete rational construction passes all729 independently checked tensor identities with23 product terms and138 nonzeros. It improves the located139-support benchmark by one; it does not reduce tensor rank or show a faster practical implementation. Its support pattern fails4 GF(2) tensor equations, so a literal sign-only scheme with this pattern is impossible. This gives a concrete...	Novelty: Promising new138-support construction; bounded independent primary search found only the authors earlier139 baseline page, not an earlier138 result. The... Formal: Complete pinned original repo retrieved and CRC/member hashes verified. Our exact checker confirms all729 tensor identities and support138/138/143, and matches every rational... M1 · D 724 · p 0.05 · 28.830 points before family union
E04-GAUGE • Gauge and inequivalence structure The gauged138 tensor is exact and shares the family credit. An independent rational-polynomial check confirms the entire support138 family for rational s outside0,-2 and specialization s=-1 to the original. Numerical Jacobian dimension, global inequivalence, minimum12constant gauges and no137boundary claims are separate statements, not implied by the finite certificates or search...	Novelty: Potentially substantive structural analysis of the new construction. The family identity is now independently checked algebraically, but its general parameter... Formal: Complete pinned original repo retrieved and CRC/member hashes verified. Our exact checker confirms all729 tensor identities and support138/138/143, and matches every rational... M1 · D 724 · p 0 · 0.000 points before family union
E04-CORE143 • Separate support-143 construction The independent exact check confirms the distinct143-support construction and its scaled Lean literals. It does not improve the138/139support frontier. Claimed different-core inequivalence and local optimality are research assertions; numerical failed searches do not prove an absolute floor.	Novelty: No eligible original mathematical delta established in the available packet. This is not a finding that all underlying work lacks novelty. Formal: Complete pinned original repo retrieved and CRC/member hashes verified. Our exact checker confirms all729 tensor identities and support138/138/143, and matches every rational... M1 · D 724 · p 0 · 0.000 points before family union

Sign-off focus: Fresh pinned kernel closure and exact hill-statement binding; Freeze-time priority/independence, including midweek public baseline exposure; Final official intake/receipt and registered family/D confirmation; Generic symmetry/inequivalence/dimension/boundary optimality claims require separate proof; no global137 impossibility

E05 • Chaewon Yoon

Shared authors: Chaewon Yoon

Proposed S 6.556 | A 6.556 | M2 0 / adjusted 0 | 10 minutes

A clean exact-arithmetic certificate collection with bounded novelty. Do not interpret public-test coverage as proof progress or combine duplicate residue coverage across teams.

Pinned commit: 9bc116919c27e63db193e03ae6b567a9f2a09579 | Repo: chaewon-research/openmath-2026

Contribution / precise assessment	Novelty / formalization / proposed score
E05-RULES • 512 residue-class certificates All512 rules passed independent exact arithmetic and cover1765/2048 odd residues mod4096. The3352/3968 public test score is a different denominator and must not be called86% of Collatz. HTPeo's234 rules yield exactly the same residue union; combining them adds nothing.	Novelty: Finite certificate collection may be new independently sealed work; the descent mechanism is classical and the union must be checked against the frozen public... Formal: Independent exact arithmetic check:512/512 valid, union1765/2048. No Lean theorem; official trust-boundary approval and mathematical novelty not established. M1 · D 772 · p 0.01 · 6.556 points before family union

Sign-off focus: Approved exact checker as formal certificate; Baseline/priority for exact rule union; No private-target coverage independently regenerated

E06 • Leanification — five-person IMPA Tech team

Shared authors: Augusto, Felipe Leria, Mateus, Tiago, Luca Seiki Pereira Fujii

Proposed S 0.000 | A 0.000 | M2 1 / adjusted 0.8 | 10 minutes

One substantial known-theorem formalization. Review semantics and fresh closure, then one M2 family. All five named contributors are retained; the.8 exception affects adjusted overall value without pretending there were.8theorems.

Pinned commit: `adcaf45c06fa70407f1abb2df80e1bd801d970b3` | Repo: `lazyluca/hyprgraph_containers`

Known-mathematics formalizations

One family per admitted source theorem or reusable formal target; supporting lemmas remain in its union.

Result ID / target	Scope / review focus
E06-THEOREM-B • Known container lemma formalization	TheoremB_unconditional plus Proposition2.2 and supporting Lemmas4.1–4.3.

Sign-off focus: Fresh pinned build and endpoint axiom audit; Independent statement/probability/algorithm fidelity review; Prior formalization search and final receipt

E08 • HTPeo — Woohyuk Kang / Hyunjin Lee / Sanghyeon Lee / Youchan Oh

Shared authors: Woohyuk Kang, Hyunjin Lee, Sanghyeon Lee, Youchan Oh

Proposed S 88.275 | A 49.421 | M2 10 / adjusted 10 | 20 minutes

Strong multi-problem packet: weighted Ramsey bound, exact Kobon construction and DMS families/reductions. M2 candidate list has 17 claimed families but seven minor/sanity entries should be held for usefulness; six explicit excluded wrappers do not count. Baseline Collatz/BB6/CHSH/matrix reconstructions earn no open points.

Pinned commit: 76c63b8e425e551a930060a6871b84fe3b4780db | Repo: lavaskiller/openmath-2026-htpeo

Contribution / precise assessment	Novelty / formalization / proposed score
E08-RAMSEY • Weighted K4 upper bound The exact weighted rational is about.030139912 rather than the McKay.030142273 reference, and is smaller than Sakana's reported bound. The generic lifting theorem matters because the finite weighted objective must imply an infinite graph-family bound. Native cross-check modules do not determine the main proof's axioms unless imported.	Novelty: Candidate exact new upper bound. Stronger prior bounds and independent sealed development still require the named audit. Formal: 2158 HTPeo formal source texts scanned, including final v2. Native.lean is explicitly separate, not imported by main claimed chain; do not reject main result for that cross-check.... M3A · D 840 · p 0.05 · 35.280 points before family union
E08-KOBON39 • 471-triangle straight-line construction The independent integer geometry recount confirms 471 bounded triangular faces for 39 lines. This improves Rohith's 470 witness by one at that order. It is a lower bound, with no proof that 471 is optimal or that the general upper 481 is attainable.	Novelty: Candidate new construction at n39; compare to historical straight-line arrangements, not just missing values in a gallery. Formal: Independent integer geometry recount gives 471, no duplicate lines. Submitted Lean existence/certificate described; fresh kernel build and source-to-geometric fidelity still... M1 · D 570 · p 0.01 · 3.574 points before family union
E08-DMS • Infinite special families and finite classification Infinite five-colour special families, finite six-colour classification up to 14 vertices, and reduction/equivalence infrastructure are materially useful. Separate unconditional endpoints from modules that retain OPEN hypotheses. Bridgeless loopless cubic multigraphs are not interchangeable with every subcubic simple graph without a proved reduction.	Novelty: Candidate obstacle-removing special-family/reduction advance; previously known five-colour classes must be subtracted. Formal: Packet provides 125-source five-colour checks and 98-module reduction library with axiom logs. An unused baseline native_decide lemma is not in claimed closure. Team declares no... M3A · D 574 · p 0.15 · 49.421 points before family union
E08-BB6 • Finite six-state baseline witness Baseline or prior known mathematics, no independently new open advance; possible useful new formalization must be separately admitted as M2.	Novelty: No eligible original mathematical delta established in the available packet. This is not a finding that all underlying work lacks novelty. Formal: Exact arithmetic certificates/source available for review; independent Collatz check 234/234. No fresh full kernel acceptance. M1 · D 974 · p 0 · 0.000 points before family union
E08-CHSH • Known CHSH witness Baseline or prior known mathematics, no independently new open advance; possible useful new formalization must be separately admitted as M2.	Novelty: No eligible original mathematical delta established in the available packet. This is not a finding that all underlying work lacks novelty. Formal: Exact arithmetic certificates/source available for review; independent Collatz check 234/234. No fresh full kernel acceptance. M1 · D 561 · p 0 · 0.000 points before family union
E08-MATRIX139 • Known support 139 tensor Baseline or prior known mathematics, no independently new open advance; possible useful new formalization must be separately admitted as M2.	Novelty: No eligible original mathematical delta established in the available packet. This is not a finding that all underlying work lacks novelty. Formal: Exact arithmetic certificates/source available for review; independent Collatz check 234/234. No fresh full kernel acceptance. M1 · D 724 · p 0 · 0.000 points before family union

Contribution / precise assessment	Novelty / formalization / proposed score
E08-COLLATZ234 • 234-rule baseline-equivalent coverage Baseline or prior known mathematics, no independently new open advance; possible useful new formalization must be separately admitted as M2.	Novelty: No eligible original mathematical delta established in the available packet. This is not a finding that all underlying work lacks novelty. Formal: Exact arithmetic certificates/source available for review; independent Collatz check234/234. No fresh full kernel acceptance. $M1 \cdot D \ 772 \cdot p \ 0 \cdot 0.000$ points before family union
E08-KOBON18 • Known $n18=93$ reconstruction Baseline or prior known mathematics, no independently new open advance; possible useful new formalization must be separately admitted as M2.	Novelty: No eligible original mathematical delta established in the available packet. This is not a finding that all underlying work lacks novelty. Formal: Exact arithmetic certificates/source available for review; independent Collatz check234/234. No fresh full kernel acceptance. $M1 \cdot D \ 570 \cdot p \ 0 \cdot 0.000$ points before family union

Known-mathematics formalizations

Ten substantive claimed families proposed; seven minor/sanity helpers held for usefulness/new-library delta; six explicitly unclaimed wrappers excluded.

Result ID / target	Scope / review focus
E08-G-PM • Petersen / Schönberger perfect matchings	<code>def Star6.Bridgeless (G : SimpleGraph V) : Prop := $\square e \square G.edgeSet, \neg G.isBridge\ e$ theorem Star6.simple_schoenberger (hconn : G.Connected) (hreg : G.IsRegularOfDegree 3) (hbr : Bridgeless G) {e : Sym2 V} (he : e $\square G.edgeSet$) : ($\square M \dots$</code>
E08-E942 • Unbounded powerful-number counts between consecutive squares	<code>@[category textbook, AMS 11] theorem erdos_942.variants.limsup : atTop.limsup (((fun (n : $\square$) $\square$ (n : $\square\infty$)) $\square$ erdos_942.h)) = $\square$</code>
E08-E44 • Greedy Sidon lower bound $N \leq A ^3$	<code>@[category textbook, AMS 5 11] theorem greedy_sidon_construction (N : $\square$) (hN : $1 \leq N$) : $\square^* (A \square Finset.lcc\ 1\ N)$, IsSidon (A : Set $\square$) $\square N \leq A.card \wedge 3$</code>
E08-E123 • d-completeness of powers of 2 and 3	<code>@[category research solved, AMS 11] theorem erdos_123.variants.powers_2_3 : IsDComplete (1(powers 2) * 1(powers 3))</code>
E08-E918 • Impossible exact countable chromatic restrictions	<code>@[category textbook, AMS 5] theorem erdos_918.variants.eq_aleph_0.parts.i : $\neg \square (V : Type\ u) (G : SimpleGraph\ V), \#V = \square_2 \square G.chromaticCardinal = \square_2 \square \square (W : Set\ V) (_ : \#W = \square_i), (G.induce\ W).chromaticCardinal = \square_o$ @[category...</code>
E08-E292 • Multiplicative closure / doubling / no prime powers	<code>@[category research solved, AMS 11] theorem erdos_292.variants.mul : $\square m \square A, \square n \square A, m * n \square A$ @[category research solved, AMS 11] theorem erdos_292.variants.two_mul : $\square n \square A, 1 < n \rightarrow 2 * n \square A$ @[category research solved, AMS 11]...</code>
E08-E395 • Radius-one reverse Littlewood–Offord counterexample	<code>@[category research solved, AMS 5 60] theorem erdos_395.variants.one : answer(False) $\leftrightarrow \square c : \square, 0 < c \square \square n : \square, 0 < n \rightarrow \square z : Fin\ n \rightarrow \square, (\square i, \ z\ i\ = 1) \rightarrow c / n \leq (signedSumCount\ z\ 1 : \square) / 2 \wedge n$</code>
E08-E698 • Erdős–Szekeres binomial gcd inequality	<code>@[category research solved, AMS 5 11] theorem erdos_698.variants.erdos_szekeres (n i j : $\square$) (hi : $1 \leq i$) (hij : $i < j$) (hj : $j \leq n / 2$) : (n.choose i : $\square$) / (j.choose i : $\square$) $\leq (Nat.gcd\ (n.choose\ i)\ (n.choose\ j) : \square) \square (2 : \square) \wedge i \leq \dots$</code>
E08-E939 • Known $r=7,8$ powerful-number sum examples	<code>@[category research solved, AMS 11] theorem erdos_939.variants.seven : (Erdos939Sums 7).Nonempty @[category research solved, AMS 11] theorem erdos_939.variants.eight : (Erdos939Sums 8).Nonempty</code>
E08-E477 • Known polynomial-range additive nonunique representation	<code>@[category research solved, AMS 12] theorem erdos_477.variants.S_sq : let! f := $X \wedge 2 \square A : Set\ \square, \square z, \neg \square! a \square A \times^* (Set.range\ f.eval), z = a.1 + a.2$ @[category research solved, AMS 12] theorem...</code>
E08-E295 • Egyptian fraction existence helper	<code>@[category textbook, AMS 5 11] lemma exists_k (N : $\square$) : $\square (k : \square) (n : Fin\ k \rightarrow \square), (\square i, N \leq n\ i) \square StrictMono\ n \square \sum i, (1 / n\ i : \square) = 1$</code>

Result ID / target	Scope / review focus
E08-E703 • Known intersecting-family $t=0$ value	@[category textbook, AMS 5] theorem erdos_703.variants.zero (n : ℕ) (hn : 1 ≤ n) : T n 0 = 2 ^ (n - 1)
E08-E748 • Known sum-free-set lower bound	@[category textbook, AMS 5 11] theorem erdos_748.variants.lower_bound (n : ℕ) : (2 : ℕ) ^ ((n : ℕ) / 2) ≤ (f n : ℕ)
E08-E1136 • Multiples-of-three density baseline	@[category research solved, AMS 11] theorem erdos_1136.variants.multiples_of_three : AvoidsPowersOfTwo {n : ℕ 3 ∤ n} ∩ Set.HasDensity {n : ℕ 3 ∤ n} (1 / 3)
E08-E358 • Sylvester consecutive sums / odd divisors	@[category textbook, AMS 5 11] theorem f_id : f id = fun n ∩ #{d ∩ n.divisors Odd d}
E08-E619 • Maximal triangle-free supergraph diameter bound	@[category research solved, AMS 5] theorem erdos_619.variants.add_edges_diam_three {V : Type*} [Fintype V] (G : SimpleGraph V) (hG : G.Connected) (hG' : G.CliqueFree 3) : ∩ H : SimpleGraph V, G ≤ H ∩ H.CliqueFree 3 ∩ H.ediam ≤ 3
E08-E1148 • Known difference-of-squares weaker bound	@[category research solved, AMS 11] theorem erdos_1148.variants.weaker : ∩ n, erdos_1148_weaker_prop n

Sign-off focus: Fresh endpoint closure build; Verify JSON→Lean identity and oracle density convention; Official receipt: report official=true, final=false; Current primary bound/independence and target D freeze; Clean Lean build, embedding/disjoint-face fidelity

E09 • Raj Harshit Srirangam

Shared authors: Raj Harshit Srirangam

Proposed S 0.000 | A 0.000 | M2 0 / adjusted 0 | 10 minutes

Most complementary Kobon upper-bound packet. Geometric soundness and exact completeness of required DP/SAT certificates are decisive. $n_{18}=93$ is not proved; all upper and finite-order claims remain one global-family union. Current proposal0 under formal-only rule; 17.8695 is a potential accepted-proof scenario, not current credit.

Pinned commit: eed14659a9b2f4ca12777da5d557f2b620b966f6 | Repo: srirangam-r/kobon_triangles

Contribution / precise assessment	Novelty / formalization / proposed score
E09-UPPER94 • Arbitrary $K(18) \leq 94$ The proposed arbitrary-multiplicity upper94 narrows n_{18} to 93 or 94. Exact DP/LP and local DRAT certificates are not enough by themselves: the geometric reduction must cover every line/pseudoline arrangement, every local view and all multiplicities. The capacity/all-8 problem still blocks upper93.	Novelty: Candidate new extension beyond simple-arrangement bounds; specific multiplicity scope is the key difference. Formal: Rational LP/DP/SAT/DRAT reports are retained, but complete geometry→constraint→certificate soundness chain has not received human or Lean review. The open93 stage is distinct from... M1 · D 570 · p 0 · 0.000 points before family union Potential accepted chain: 17.8695
E09-TRIPLES93 • Multiplicity ≤ 3 upper93 Finite upper bounds/classification are partial in all-n family, not $p=1$ for global target. All overlapping n_{18} reductions and restricted results union once. Stage2 fraction of cubes is not logical progress.	Novelty: Candidate scoped mathematical delta; exact baseline and independently sealed provenance are required. A missing gallery value is not evidence of a world record. Formal: Rational LP/DP/SAT/DRAT reports are retained, but complete geometry→constraint→certificate soundness chain has not received human or Lean review. The open93 stage is distinct from... M1 · D 570 · p 0 · 0.000 points before family union Potential accepted chain: 17.8695
E09-K14 • Claimed complete finite n_{14} case $K_{14}=54$ is a claimed complete finite child result under arbitrary multiplicities. In this review it remains within the same general Kobon family as the upper18 and even-order bounds. Re-registering every solved finite order to stack full $p=1$ scores would inflate the family union.	Novelty: Candidate new arbitrary-multiplicity closure, subject to actual historical scope and certificate coverage. Formal: Rational LP/DP/SAT/DRAT reports are retained, but complete geometry→constraint→certificate soundness chain has not received human or Lean review. The open93 stage is distinct from... M1 · D 570 · p 0 · 0.000 points before family union Potential accepted chain: 17.8695
E09-EVEN • Triple-point bounds across even n Finite upper bounds/classification are partial in all-n family, not $p=1$ for global target. All overlapping n_{18} reductions and restricted results union once. Stage2 fraction of cubes is not logical progress.	Novelty: Candidate scoped mathematical delta; exact baseline and independently sealed provenance are required. A missing gallery value is not evidence of a world record. Formal: Rational LP/DP/SAT/DRAT reports are retained, but complete geometry→constraint→certificate soundness chain has not received human or Lean review. The open93 stage is distinct from... M1 · D 570 · p 0 · 0.000 points before family union Potential accepted chain: 17.8695
E09-ALL8 • Residual $K_{18}=93$ obstruction Finite upper bounds/classification are partial in all-n family, not $p=1$ for global target. All overlapping n_{18} reductions and restricted results union once. Stage2 fraction of cubes is not logical progress.	Novelty: Candidate scoped mathematical delta; exact baseline and independently sealed provenance are required. A missing gallery value is not evidence of a world record. Formal: Rational LP/DP/SAT/DRAT reports are retained, but complete geometry→constraint→certificate soundness chain has not received human or Lean review. The open93 stage is distinct from... M1 · D 570 · p 0 · 0.000 points before family union Potential accepted chain: 3.5739
E09-K10 • Known K_{10} / n_{18} construction Known prior values; no open-problem credit.	Novelty: No eligible original mathematical delta established in the available packet. This is not a finding that all underlying work lacks novelty. Formal: Certificates may be reusable research evidence. M1 · D 570 · p 0 · 0.000 points before family union Potential accepted chain: 0.0

Sign-off focus: Verify all geometric transfer/local-redrawing lemmas; Check exact rational LP and DP certificates and DRAT proof-to-model chain; Historical n_{14} /triple-point scope and target registration; M2 contribution only if new useful formalization

E10 • Jamie Steeg / grandchallenge / GCL

Shared authors: Jamie Steeg

Proposed S 6.556 | A 6.556 | M2 1 / adjusted 1 | 10 minutes

Only the bounded Collatz classification currently merits conditional open credit. Kobon predecessor assumptions are not established. AP conclusion bridges are one M2 candidate; official fallback receipt remains an operations question.

Pinned commit: 2fd9fd643bf3e9e98d121dc2e7af1cf1be411ae2 | Repo: grandchallenge/MATHSOLVE

Contribution / precise assessment	Novelty / formalization / proposed score
E10-H4 • 23-rule bounded classification The 23-rule union independently covers 1504 of the 1765 classes already covered by Chaewon/HTPeo. The bounded depth ≤ 4 and $k \leq 8$ classification may be a distinct method result, but its examples do not expand the collective convergence region.	Novelty: Potential new bounded-classification formal result; no new collective residue coverage. Formal: Exact integer certificate passes our check; 94 odd classes mod 256, all inside Chaewon/HTPeo union. Full 309/1016 catalogue enumeration not independently replayed; no official... M1 · D 772 · p 0.01 · 6.556 points before family union
E10-H1PART • Conditional hypothetical 95 exclusion The hypothetical 95 exclusion retains predecessor local-fan and clean-line assumptions, and $q \geq 6$ remains open. An implication from unproved geometric premises does not establish an unconditional Kobon improvement.	Novelty: Conditional organization of the obstruction may be useful, but no completed eligible original advance is established. Formal: Predecessor facts/return evidence and replay are not all independently discharged. M1 · D 570 · p 0 · 0.000 points before family union
E10-H1BASE • Known Bader 93 Prior benchmark only; no new open advance identified.	Novelty: No eligible original mathematical delta established in the available packet. This is not a finding that all underlying work lacks novelty. Formal: Queued native reports and prepared packet are not official final receipts. M1 · D 570 · p 0 · 0.000 points before family union
E10-H2 • Finite baseline witness Prior benchmark only; no new open advance identified.	Novelty: No eligible original mathematical delta established in the available packet. This is not a finding that all underlying work lacks novelty. Formal: Queued native reports and prepared packet are not official final receipts. M1 · D 974 · p 0 · 0.000 points before family union
E10-H5 • CHSH baseline Prior benchmark only; no new open advance identified.	Novelty: No eligible original mathematical delta established in the available packet. This is not a finding that all underlying work lacks novelty. Formal: Queued native reports and prepared packet are not official final receipts. M1 · D 561 · p 0 · 0.000 points before family union
E10-H6 • Laderman baseline Prior benchmark only; no new open advance identified.	Novelty: No eligible original mathematical delta established in the available packet. This is not a finding that all underlying work lacks novelty. Formal: Queued native reports and prepared packet are not official final receipts. M1 · D 724 · p 0 · 0.000 points before family union

Known-mathematics formalizations

One family per admitted source theorem or reusable formal target; supporting lemmas remain in its union.

Result ID / target	Scope / review focus
E10-H7 • AP conclusion representation bridge	ap_subprogression_of_le; ap_frequently_iff_every_length.
Sign-off focus: Novelty and approved checker; Official route remains pending; email/repo received $\neq$ accepted; Do not use later adcfa053 mathematical changes; Scoped prior formal-library search/usefulness; Pinned image semantic fidelity and final intake	

E11 • Leon Koerbs / Cameron Fen — Erdős #3 Token Burner

Shared authors: Leon Koerbs, Cameron Fen

Proposed S 0.000 | A 0.000 | M2 3 / adjusted 3 | 10 minutes

Formal and analytic Erdős groundwork, not a solution. Deduplicate imported known3-term work, dyadic equivalence and Gowers machinery. The finite harmonic4.314 is below Walker's known4.43975 and Sakana's claimed4.44.

Pinned commit: 551e388b0a7a933e68caddbab099342af326c1dc2 | Repo: Koerbser/erdos-3-autolab

Contribution / precise assessment	Novelty / formalization / proposed score
E11-HARMONIC • Finite four-AP-free harmonic construction Below packet-cited existing infinite4.43975; no new bound on f(4) established. FiniteN benchmark relevance needs its own admitted target/D, not borrowing parentD.	Novelty: No eligible original mathematical delta established in the available packet. This is not a finding that all underlying work lacks novelty. Formal: Hill integer/bitmap checks reported; finite harmonic sum not a divergent series. M3A · D 538 · p 0 · 0.000 points before family union

Known-mathematics formalizations

One family per admitted source theorem or reusable formal target; supporting lemmas remain in its union.

Result ID / target	Scope / review focus
E11-THREE • Known length-three case	Formal divergent-reciprocal implication for3-term APs using external Kelley–Meka formalization.
E11-DYADIC • Per-length extremal equivalence	For every $K \geq 3$, dyadic $r_K(2^j)/2^j$ summability iff every divergent set contains K -term AP.
E11-GOWERS • Four-term analytic groundwork	norm_apCount4_pow8_le_gowers3 and associated closed counterexample-audit lemmas.

Sign-off focus: Immutable complete dependency closure/license; Prior formalization and wrapper usefulness; Fresh full check; Fresh named endpoint + imported Dyadic closure; Check gluing uses sparse8^jwindows and handles $K \geq 3$

E12 • Frederik Smits van Oyen — K4 Ramsey and Heilbronn

Shared authors: Frederik Smits van Oyen

Proposed S 0.000 | A 0.000 | M2 0 / adjusted 0 | 10 minutes

Large Heilbronn dataset, incomplete global cover. Sampling and missing leaves prevent a global theorem; classify and preserve local artifacts but recommend no original-family points yet.

Pinned commit: b8da072acb21d2b0acc8daa038600dddfc8c67e9 | Repo: FSvOAI/Heilbronn

Contribution / precise assessment	Novelty / formalization / proposed score
E12-HEILBRONN • Incomplete nine-point cover The target value is about.0274262, but the global box cover has2581 unresolved and529 missing leaves, with boundary coverage incomplete. A large number of solved boxes cannot prove a universal global bound. Preserve successful local bounds as reusable computational evidence.	Novelty: Potential partial covering/certificate engineering; no accepted global theorem or complete new bound established. Formal: Author floating-bound audit uses tolerance/protocol; no complete formal global certificate. M3B · D 715 · p 0 · 0.000 points before family union
Sign-off focus: Domain, boundary and missing-leaf coverage; Approved exact outward-bound arithmetic; Separately specified certified box subclaims if proposed; Final exact artifact and immutable on-time designation; Approved checker/semantic review	

E17 • Qichao Wang — Černý partial formalizations

Shared authors: Qichao Wang

Proposed S 0.000 | A 0.000 | M2 0 / adjusted 0 | 10 minutes

Detailed Černý declarations identify genuinely limited results; no general upper-bound improvement. Original27MBsource is absent. Recommend five reusable M2 groups from nine claimed sections, all held pending source, rather than counting100declarations.

Pinned commit: ac948e7001bff5428c22d236f96974856452d34d | Repo: None

Contribution / precise assessment	Novelty / formalization / proposed score
E17-PAPER • Unformalized all-parameter families The paper-to-concrete-automaton and all-parameter bridges are not supplied in a reproducible formal packet. Abstract lower bounds cannot prove the all-n Černý upper bound. The 27MB original source is missing despite detailed manifests.	Novelty: Potential method-obstruction and lower-bound research; no improved general reset upper bound is claimed. Formal: Original paper/proofs/source absent locally; exact abstract components do not certify concrete parameterized automata. M3A · D 855 · p 0 · 0.000 points before family union

Known-mathematics formalizations

Five groups: generic potential/research/layered framework(CORE1,2,6); anchor/count-window(CORE3,4); C3/C4 finite optimality(CORE5); Kari shrink/phase(CORE7,8); reflection finite-depth(CORE9). All held because original source is missing. Current proposal0, potential5 after original source delivery and formal-novelty review.

Result ID / target	Scope / review focus
E17-CORE1 • `Orion.potential_along_word`, `Orion.lower_bound`, `Orion.cardinality_limit`	For arbitrary step systems, local potential inequalities telescope along every list; a zero-potential goal forces word length $\geq$ initial potential. The cardinality-slope lemma is conditional on its stated slope assumptions.
E17-CORE2 • `OrionResearch.potential_lower_bound`, `OrionResearch.quotient_local`,...	Arbitrary relational walks and existential quotient-edge local transfer; inclusion-type monotone simulation; finite dyadic timing lower bounds, optimal-tail rigidity and buffer arithmetic; eventual constancy of decreasing natural...
E17-CORE3 • `OrionAnchors.p2_rotation`, `OrionAnchors.p2_merge`, `OrionAnchors.p3_rotation`, `OrionAnchors.p3_merge`,...	Symbolic two-/three-anchor rotation and merge inequalities for all valid cardinalities and bits, and the full three-anchor potential value for $n \geq 5$.
E17-CORE4 • `OrionThreeGraph.one_step`, `OrionThreeGraph.full_lower_bound`	In the explicitly defined count/window graph, any walk from the full feature to a cardinality-one goal has length $\geq 3n-4$, $n \geq 5$. This is a lower bound in that graph, not the general exact quotient formula.
E17-CORE5 • `OrionExample.one_step_fidelity`, `OrionExample.optimal_subset_path` (separate namespaces in separate...	In the separately defined finite C3 and C4 powerset tables, the listed witnesses attain the singleton goal and every goal-reaching list is at least as long: lengths 4 and 9 respectively. Every table edge has a checked one-step...
E17-CORE6 • `Orion.DirectAttack.layered_invariants`, `Orion.DirectAttack.first_collapse_cost`	For any `LayeredCertificate` satisfying all local index, internal/entrance-potential and K/U entrance hypotheses, the five invariants hold along every path. A path ending in K or U followed by its collapse letter costs at least the...
E17-CORE7 • `KariInverseShrink.literal_preimage_correct`, `KariInverseShrink.no_monotone_inverse_reset`,...	For the literal six-state preimage-mask system, no cardinality-nondecreasing inverse word takes any singleton to the full mask. Every inverse word reaching full has ≥ 1 decreasing step; a specific length-25 inverse word from singleton4...
E17-CORE8 • `KariProductBase.reset_length_lower`, `KariProductBase.to45_length_lower`,...	In that same literal base system, every singleton-to-full inverse word has length ≥ 25 . The last decreasing edge is $45 \rightarrow 44$ or $47 \rightarrow 60$; a sole decrease is $47 \rightarrow 60$. The stated prefix/monotone-suffix hypotheses at $45/44$ force cost ≥ 27 ; phase...
E17-CORE9 • `ReflectionCoreCut.cut_closed`, `ReflectionCoreCut.run_in_layers`, `ReflectionCoreCut.no_avoid_first_eight`,...	In the abstract integer-hole recurrence, checked layers close through eight steps; every word of length ≤ 8 has `SafeRun`. The specified eight-letter prefix ends with holes [0,1,3], which are unsafe.

Sign-off focus: Actual sourceZIP and exact on-time revision; Prior status and family deduplication; Abstract quotient→concrete DFA bridge, explicitly absent where claimed; Full paper/source and approved formal bridge; Event delta vs imported baseline

E57 • Rohith Poola — Kobon only

Shared authors: Rohith Poola

Proposed S 3.574 | A 3.574 | M2 0 / adjusted 0 | 10 minutes

Exact66-arrangement catalogue is independently recounted. Forty-six rows marked NEW by missing gallery values are not automatically world records. Score one parent construction family if historical novelty and eligible formal certificate protocol are confirmed.

Pinned commit: f462d8e18aea2a458376c523c9b6c2980237071f | Repo: Rohith18p/rsi-kobon-triangles

All66 straight-line witnesses were independently recounted and match their submitted triangle counts. Forty-six catalogue entries are marked NEW by the author; missing gallery values are not established world records. n39=470 is superseded here by HTPeo471; n18=93 ties known mathematics. One P1 parent family, not66 awards. The generalized conjecture and unrealized pseudoline optima receive0.

n : triangles	n : triangles	n : triangles	n : triangles
7 lines/11triangles	13 lines/47triangles	18 lines/93triangles	19 lines/107triangles
31 lines/290triangles	35 lines/338triangles	37 lines/431triangles	38 lines/450triangles
39 lines/470triangles •candidate	40 lines/494triangles •candidate	41 lines/533triangles	43 lines/587triangles
44 lines/608triangles •candidate	45 lines/645triangles	46 lines/667triangles	47 lines/691triangles •candidate
48 lines/721triangles •candidate	49 lines/767triangles	50 lines/792triangles	51 lines/815triangles •candidate
52 lines/850triangles •candidate	53 lines/901triangles	54 lines/927triangles	55 lines/954triangles •candidate
56 lines/990triangles •candidate	57 lines/1045triangles	58 lines/1036triangles •candidate	59 lines/1086triangles •candidate
60 lines/1137triangles •candidate	61 lines/1190triangles •candidate	62 lines/1185triangles •candidate	63 lines/1243triangles •candidate
64 lines/1302triangles •candidate	65 lines/1365triangles	66 lines/1329triangles •candidate	67 lines/1383triangles •candidate
68 lines/1438triangles •candidate	69 lines/1494triangles •candidate	70 lines/1525triangles •candidate	71 lines/1590triangles •candidate
72 lines/1657triangles •candidate	73 lines/1727triangles	74 lines/1438triangles •candidate	75 lines/1489triangles •candidate
76 lines/1540triangles •candidate	77 lines/1595triangles •candidate	78 lines/1650triangles •candidate	79 lines/1708triangles •candidate
80 lines/1767triangles •candidate	81 lines/1829triangles •candidate	82 lines/1891triangles •candidate	83 lines/1956triangles •candidate
84 lines/2021triangles •candidate	85 lines/2090triangles •candidate	86 lines/2159triangles •candidate	87 lines/2233triangles •candidate
88 lines/2307triangles •candidate	89 lines/2384triangles •candidate	90 lines/2461triangles •candidate	91 lines/2542triangles •candidate
92 lines/2623triangles •candidate	93 lines/2709triangles •candidate	94 lines/2795triangles •candidate	95 lines/2885triangles •candidate
96 lines/2977triangles •candidate	97 lines/3071triangles		

Each witness has its own entries/E57-K<n>.txt assessment and file summary. Candidate labels are proposed novelty holds; all entries deduplicate to3.5739 parent points.

Sign-off focus: Historical priority/frozen lower-bound table; Approved exact geometric checker; Shared gallery source attribution; Prove exact conjecture / straightening if later research

E65 • Wilson Wu — Erdős supporting formalizations

Shared authors: Wilson Wu

Proposed S 0.000 | A 0.000 | M2 2 / adjusted 2 | 10 minutes

Useful closed Erdős equivalence and AP bridge infrastructure. Mathematical claims are appropriately conditional. Conservatively propose two M2 families, not a new open-conjecture score. Compare exact overlap and sealed provenance with E11 and E10.

Pinned commit: 492fd601f9c76e44dcb87f683a5ea6ccb7aaefc9 | Repo: wilsonwu-ai/sundai-erdos-3

Known-mathematics formalizations

Two groups: dyadic extremal equivalence; AP interface/conditional analytic transfer. The conditional three-term bridge is not a separate proved analytic theorem.

Result ID / target	Scope / review focus
E65-DYADIC • Exact hill/extremal equivalence	Original hill iff $\sum_j r_k(2^j)/2^j$ converges for every $k \geq 4$; neither side proved.
E65-THREE • Conditional three-term analytic bridges	Kelley–Meka/Bloom–Sisask density hypotheses imply known three-term target; analytic bounds are not proved here.
E65-ELEMENTARY • Elementary AP vocabulary / special cases	Cofinite sets, affine tails, unboundedness→two-term, AP truncation and library bridges.

Sign-off focus: Fresh endpoint closure/kernel and source-definition fidelity; Independence/priority against Leon equivalence; Formal novelty/usefulness; Scoped new formalization delta; Hypotheses printed and analytic result provenance

Problem-centered scientific frontier

Read these as virtual problem entrants: all partial and complete contributions are assembled together, with the same family-union rule. Their numbers are analytical references, not extra awards. A complementary result does not automatically raise p; it must close an additional logical obstacle.

KOBON • OPDP D 570

Frontier: Lower constructions: independently recounted HTPeo $K(39) \geq 471$, Rohith $K(39) \geq 470$ and 66 catalogue witnesses. Raj claims arbitrary-multiplicity $K(18) \leq 94$, restricted ≤ 93 , $K(14) = 54$ and triple-point even-order bounds; the formal geometry transfer is incomplete. Jamie conditional exclusions do not establish an unconditional bound.

Interaction: At each SAME n, compare a straight-line lower witness to an upper bound with matching degeneracy assumptions. n39 lower and n18 upper do not numerically close one another. Normalize wiring words, line coordinates, multiplicities and bounded faces; pseudoline lower witnesses require realizability.

Collective: Conservative virtual entrant: one current parent-family $P1 = .01$ (3.5739 points with focus bonus); no sum of the 66 arrangements. Potential $P2 = .05 / 17.8695$ only after the upper-bound proof chain is accepted.

Paper: An exact-construction catalogue with independent recounts is plausible now. A joint construction/upper-bound paper is conditional on Raj closing the geometric soundness chain. Attribute pre-existing 93/18 witnesses and imported gallery configurations; give authors credit by actual work, not leaderboard presence.

COLLATZ • OPDP D 772

Frontier: E05 and E08 cover exactly the same 1765 odd residue classes modulo 4096; E10 covers 1504 classes, all within that union. E66 lacks a complete inspectable packet. This is partial certified descent, not all-integer Collatz.

Interaction: Compare the depth/descent inequalities and canonical rule basis, not counts of rules or private weighted oracle scores. Compression and bounded-classification lemmas may differ while convergence coverage is identical.

Collective: Virtual family credit stays $p = .01$, 6.555824 with the focus bonus; duplicate/subset coverage adds zero. Individual sealed-provenance credit is a separate eligibility decision.

Paper: One shared exact checker/coverage paper is more credible than three papers claiming the same region. Include independent provenance and genuinely different bounded-classification or compression methods.

RAMSEY-K4 • OPDP D 840

Frontier: HTPeo exact bound $\sim .030139912$ is stronger than Sakana $\sim .0301409323$ and the located McKay reference $.0301422734319$. E01 and several account leads lack graphs/weights. No lower bound matching the upper is supplied.

Interaction: Reuse a faithful generic weighted-template lifting theorem to validate every exact graph packet under one density convention. Rank exact rationals after validation; board values alone cannot establish an infinite theorem.

Collective: One conservative $P2 = .05$ frontier, 35.28 under proposed D840 and conservative $b = 1$. If inclusion in the frozen focus set is confirmed, $b = 1.1$ gives 38.808.

Paper: A strongest verified weighted construction with a generic formal lifting library is promising. Verify the frozen reference, stronger prior literature and sealed independence before attributing novelty.

MATRIX-3 • OPDP D 724

Frontier: Chandra complete pinned original is now available. Our checker confirms all 729 rational tensor identities for support 138, its gauged 138 version and core 1809 support 143, plus exact scaled Lean-data correspondence. The rational one-parameter 138 family is independently checked outside $s = 0, -2$. Sakana one-leg rigidity for known 139 seeds is narrower.

Interaction: Use the verified 138 tensor as a new seed for restricted one-leg rigidity and compare the same support/gauge conventions. The literal 138 support pattern fails 4GF(2) identities; sign-only search cannot produce this pattern. Fixed-factor rigidity still cannot prove global 137 impossibility.

Collective: Conservative one-family $P2 = .05$ at D724 gives 28.82968, pending fresh closure and priority/intake. Sakana $P1 = .01$ gives 5.765936 individually; their scientific frontier does not sum these overlapping family points.

Paper: Strong candidate: exact 138 construction plus rational family and restricted rigidity. Constant arithmetic/runtime claims, global inequivalence, dimension and boundary minimality need separate proofs. Attribute source-specific work and repair stale README scale-factor entries.

ERDOS3 • OPDP D 674

Frontier: Leon/Cameron and Wilson supply overlapping extremal/dyadic equivalences; Jamie a conclusion bridge. Three-term Kelley–Meka and Gowers 4 material is known mathematics. Neither side of the all-length equivalence is discharged. Sakana proves a mod 13 diagonal-pullback obstruction; its 4-AP sum > 4.44 belongs to Erdős 169, not a solution here.

Interaction: Normalize progression length, density/maxCard and dyadic summability definitions. Align exact endpoints and imported dependencies; preserve conditional analytic hypotheses.

Collective: No full solution. Sakana narrow method-obstruction proposal $p=.01$ gives 4.997036; known equivalences earn no original-result points. Proposed M2 frontier families: dyadic equivalence, AP/conclusion interface, three-term integration, Gowers4 estimate. Separate implementation novelty from mathematical novelty.

Paper: A reusable formal interface joining extremal counting and reciprocal-sum conclusions is plausible. Keep the all-length density estimate clearly open; deduplicate the equivalence while recording independent implementations.

ERDOS169 • OPDP D 538

Frontier: Sakana finite4-AP-free positive set has harmonic sum >4.44 ; Leon/Cameron report 4.314 in a fixed-N benchmark. The located prior numerical claim 4.4397534742 is smaller than 4.44.

Interaction: Compare exact same-domain reciprocal sums; N-dependent benchmarks require matching size/domain. Unbounded harmonic sums do not follow from these finite examples.

Collective: One original-family $P2=.05$ frontier at D538:14.4722, not two independent summed scientific bounds.

Paper: A formal finite construction and reusable AP-free certificate method may merit a short paper. Preserve lower-scoring fixed-N work only with a distinct algorithmic/formal contribution.

DMS • OPDP D 574

Frontier: HTPeo gives 125-source special-family five-colour work and 98-module reductions, finite six-colour classification up to 14 vertices and endpoints retaining OPEN hypotheses. Not every subcubic graph is covered.

Interaction: Petersen/perfect-matching infrastructure supports concrete reductions. Container formalization is a prospective combinatorics resource, with no demonstrated DMS consequence this event.

Collective: Conservative proposed $P3=.15$ at D574:49.4214; fallback $P2=.05$ gives 16.4738 if only the narrower new scope is admitted.

Paper: A formal special-family/reduction library is plausible if claimed novelty survives known-class comparison. Separate original endpoints, known theorem formalization and still-open general lifting.

BB6 • OPDP D 974

Frontier: Available finite six-state halting traces tie or trail known records; board ties can come from different machines. Sakana reports 1,235,211 steps/1519 ones, not the busy-beaver maximum. Tissue/adaptive models are different targets.

Interaction: Normalize machine tables, blank tape, head position, step counting, halting/ones/span conventions. Share bounded trace checking; do not merge machines from equal scores.

Collective: Original-family proposal 0. Preserve partial traces and possible reusable M2 targets, without inventing theorem credit.

Paper: A generic formal bounded-machine certificate library could be useful; a numerical subrecord needs a separate method novelty claim.

GROTHENDIECK • OPDP D 561

Frontier: CHSH and tiny finite sign enumerations are known/baseline results. Sakana proves a three-row square-root 1.5 bound and a four-point square-root 2 enumeration; the universal constant remains untouched.

Interaction: Classify restricted dimensions/sign domains and compare exact finite certificates. A restricted proof is not a universal Grothendieck improvement.

Collective: Original-family 0; known restricted formalization credit must clear baseline overlap before M2 admission.

Paper: Only a distinctly new reusable restricted formal theorem might justify a library note; current packets do not support a new universal-constant paper.

CERNY • OPDP D 855

Frontier: Qichao reports nine abstract core sections and a research paper. The original 27MB source packet is absent; generic potential/layered and inverse-shrink interfaces do not prove the all-n reset bound.

Interaction: Deduplicate to five coherent M2 groups; prove a literal transition-system bridge and all-parameter hypotheses before claiming Černý progress.

Collective: Original-family 0. Five M2 groups are potential proposals on hold, not verified delivered formalizations.

Paper: A scoped method-obstruction/interface paper is possible only after the actual source is retrieved and checked; no general conjecture-resolution paper.

HEILBRONN9 • OPDP D 715

Frontier: Nine points in the reference triangle T; attempted target .027426211734693878. Coverage has 2581 unresolved and 529 missing leaves, plus incomplete boundary coverage. Local solver PROVED labels are not a global proof.

Interaction: Unify canonical box IDs and split trees, handle corners/boundary, use outward-rounded certified bounds and prove cover completeness. Preserve successful local subcases.

Collective: Current original-family 0; percentages of solved boxes do not determine p.

Paper: Currently a reproducible computational partial-coverage report, not a universal nine-point theorem. Publication of a global bound requires every leaf closed or covered.

UNITARY • OPDP D 824

Frontier: Matt proposes asymptotic $21/48 \cdot 4^n$ versus the located published $22/48$ baseline. The n^3 circuit experiments do not prove exact surjectivity for every unitary, and the inductive manuscript is nonrigorous.

Interaction: Bridge exact circuit construction, worst-case completeness and uniform induction; numerical synthesis examples are useful experiments only.

Collective: Current formal-only; no score from the claimed coefficient alone.

Paper: A methods/experimental note is possible; a new worst-case bound awaits an actual proof and formalized universal construction.

Other original targets

OEIS-A100475 | M3A • D 541 • p 1 • 292.681 points before family union

The positive-start restriction is essential. Growth of the prime-index map plus finite exceptional checks can rule out periodic tails for every positive start. Zero-start totalization would answer a different question. This is a complete target candidate, with a substantial source chain for a deceptively short statement. Novelty: Candidate original resolution; bounded searches found no exact positive-start solution. Ball–Coxeter and the 2004 SeqFan discussion remain specific priority checks, so novelty is...

Publication relation: Standalone scoped result or related-family chapter, contingent on exact prior comparison and formal closure. Full scope/inputs/source analysis: problems/OEIS-A100475.txt.

OEIS-A060957 | M3A • D 527 • p 1 • 277.729 points before family union

This refutes the actual whole-interval distinct-subset-product interpolation assertion, not an arbitrary restricted factor set. The arbitrary-gap theorem is materially stronger but stays in this family. Prime-coordinate encoding and the exact eight-level fibre are the useful reusable mechanism. Novelty: Strong original-result candidate against the 2020 OEIS conjecture; the located prior work supplies methods, not this complete-interval counterexample. First-publication priority...

Publication relation: A family paper can include its quantitative strengthening without extra family credit. Full scope/inputs/source analysis: problems/OEIS-A060957.txt.

OEIS-A000224 | M3A • D 622 • p 0.15 • 58.033 points before family union

The uniform product-of-three-distinct-odd-primes theorem removes an infinite subcase. It does not handle repeated prime powers, four or more factors, or the entire OEIS equivalence. The exact definition counts zero as a square residue. Novelty: Plausible new infinite-family advance; the located prior Pell scan is bounded and does not imply this theorem.

Publication relation: Standalone scoped result or related-family chapter, contingent on exact prior comparison and formal closure. Full scope/inputs/source analysis: problems/OEIS-A000224.txt.

LATIN-TABLEAU | M3A • D 650 • p 0.15 • 63.375 points before family union

Preservation of an arbitrary supplied optimal factor is stronger than mere existence of one convenient factor. The hierarchy determines later matching-union ranks, while the $e > 1$ case does not realize every individual early CDS colour. That last limitation prevents a full Latin-tableau conclusion. Novelty: Candidate new structural extension beyond known boundary cases. Some factor-theory full texts remain an explicit literature gap.

Publication relation: Standalone scoped result or related-family chapter, contingent on exact prior comparison and formal closure. Full scope/inputs/source analysis: problems/LATIN-TABLEAU.txt.

OPDP13-QUARTIC | M3B • D 644 • p 1 • 207.368 points before family union

The precise quartic family has both an exact exponent-semigroup classification and a matching asymptotic threshold $\delta^* N / \log(1/\delta)$ tending to one. The source uses a function-level holomorphic squared-norm representation, which is stronger than a coefficient-only claim. Novelty: Original variation candidate. Tan–To’s inspected appendix concerns multiplication by R^m , a different operation, but its main theorem and Handelman full text still need the named...

Publication relation: A family paper can include its quantitative strengthening without extra family credit. Full scope/inputs/source analysis: problems/OPDP13-QUARTIC.txt.

NO3-HALFTURN | M3A • D 330 • p 1 • 108.900 points before family union

A $2n$ -point no-three-collinear grid set has exactly two points per row and column. The incidence discrepancy with its half-turn image has zero row and column balances, so a nonempty discrepancy must include an alternating cycle and at least four one-sided misses. The ten-point 5×5 witness attains four; symmetric difference is eight. Novelty: The near-symmetry question is present on the pre-event Bielefeld source. The argument is short and has material folklore risk, but that is not proof of prior publication.

Publication relation: Standalone scoped result or related-family chapter, contingent on exact prior comparison and formal closure. Full scope/inputs/source analysis: problems/NO3-HALFTURN.txt.

ERDOS1060 | M3A • D 574 • p 0.05 • 16.474 points before family union

The proposed uniform exponent constant is about.363272, below $\log(3)/3$ about.366204. The separately located odd-target constant $\log(2)/2$ about.346574 applies only to odd n and does not subsume this all-target theorem. A positive constant still misses the requested little-o exponent. Novelty: Candidate new all-target refinement; isolate the all- n domain from prior odd-only bounds and known powerful-core machinery.

Publication relation: Standalone scoped result or related-family chapter, contingent on exact prior comparison and formal closure. Full scope/inputs/source analysis: problems/ERDOS1060.txt.

ERDOS21-VARIANT | M3A · D 555 · p 1 · 308.025 points before family union

The +3 residual question is explicitly asked in the July2026 paper, so it predates the competition. Closing it changes the elementary all- r bound from $3r-4$ to $3r-3$. The source's stronger approximately $3.053r$ asymptotic bound is not improved. Imported matching infrastructure must remain attributed. Novelty: Strong explicit-subquestion candidate: the inspected primary paper asks for +3 after its +4 lemma. The known triangle sharpness example and imported author prefix are supporting...

Publication relation: Standalone scoped result or related-family chapter, contingent on exact prior comparison and formal closure. Full scope/inputs/source analysis: problems/ERDOS21-VARIANT.txt.

ERDOS944 | M3A · D 721 · p 0.05 · 25.992 points before family union

The linear constraint $n \geq (27r+19)/4$ beats $5r+6$ for integer $r \geq 1$ after rounding, tying at $r=1$. The source's $n/(\log n)^c$ asymptotic restriction remains stronger eventually. This is a useful explicit finite-regime necessary condition, not a new construction or full existence answer. Novelty: Candidate finite-regime improvement over the explicit prior bound. Existing $r=1$ graph constructions and classical coding inequalities receive no new-mathematics credit.

Publication relation: Standalone scoped result or related-family chapter, contingent on exact prior comparison and formal closure. Full scope/inputs/source analysis: problems/ERDOS944.txt.

RBM4-PARITY | M3B · D 727 · p 1 · 264.264 points before family union

The theorem concerns the full nine-dimensional four-visible/one-hidden RBM, not only a one-dimensional ray. The strict threshold region has non-strict local maxima; equality and beyond are saddles. Every collapsed point is still suboptimal globally. The degenerate higher-order analysis is the substantive contribution. Novelty: Candidate exact new threshold; collapsed stationarity, singular mixture lines and interaction formulas are prior work. No claim for general training landscapes or $n \geq 5$ is...

Publication relation: A family paper can include its quantitative strengthening without extra family credit. Full scope/inputs/source analysis: problems/RBM4-PARITY.txt.

ERDOS829 | M3A · D 530 · p 0.05 · 14.045 points before family union

The all-target bound uses coefficient $\log(5/3)$, about.510826, rather than the earlier same-family $\log(7/4)$, about.559616. The located.556477 comparison is only cubefree and does not cover all targets. Including zero coordinates and ordered pairs preserves the original count. The existential threshold is not effective. Novelty: Candidate new uniform bound built from prior ratio-root, determinant and coding methods. Earlier intermediate coefficients are revisions, not additional families.

Publication relation: Standalone scoped result or related-family chapter, contingent on exact prior comparison and formal closure. Full scope/inputs/source analysis: problems/ERDOS829.txt.

Every other record retained

These records currently have 0 original points and 0 M2. No final inspectable mathematical packet was available, or the account expressed interest/withdrew. This is not a finding that unexposed/private work is mathematically worthless. All source identities, aliases and individual result rows remain in the local library.

Entry / identity	Disposition / source status	Result IDs
E01 Luke Van Seters / Madhan	Promising score / final packet missing	E01-RAMSEY
E07 Chris Forrester / Hypernym	Project leads / incomplete or private artifacts	E07-RAMTASTIC, E07-BOSONIC
E13 Ilknur Icke — Tissue Busy Beaver	Exploratory computational-biology interest; no formal competition packet claimed	E13-RESEARCH
E14 Willem Nielsen — Adaptive Turing Machines	Project only / no final packet	E14-RESEARCH
E15 Shafeeq — Ramsay Multiplier K4	Ramsey scored run; named public repo is empty of result material	E15-RAMSEY
E16 Chenguang Xu	CLOSED — author confirms no work and no participation this time	E16-NO-RESULT
E18 Suraj Saxena	Registered candidate contacted; no overnight final artifact found	E18-NO-RESULT
E19 Will	Registered candidate contacted; no overnight final artifact found	E19-NO-RESULT
E20 Mick Darling	Registered candidate contacted; no overnight final artifact found	E20-NO-RESULT
E21 Ganesh Prakash Jamadar	Registered candidate contacted; no overnight final artifact found	E21-NO-RESULT
E22 Krish Vijay	Registered candidate contacted; no overnight final artifact found	E22-NO-RESULT
E23 Kevin Srun	Registered candidate contacted; no overnight final artifact found	E23-NO-RESULT
E24 Thinking Machines — Jonathan Duran / Rishik Pavani / Zachary Norton	Registered candidate contacted; no overnight final artifact found	E24-NO-RESULT
E25 Aaron Soetopo	Registered candidate contacted; no overnight final artifact found	E25-NO-RESULT
E26 Young Jae Koh	Registered candidate contacted; no overnight final artifact found	E26-NO-RESULT
E27 Yev Barkalov	Registered candidate contacted; no overnight final artifact found	E27-NO-RESULT
E28 Stan / Stanislav Srednyak	Registered candidate contacted; no overnight final artifact found	E28-NO-RESULT
E29 Antoine de Saint Germain	Registered candidate contacted; no overnight final artifact found	E29-NO-RESULT
E30 Ryan Bahadori	Registered candidate contacted; no overnight final artifact found	E30-NO-RESULT
E31 Yikun Ding	Registered candidate contacted; no overnight final artifact found	E31-NO-RESULT
E32 Sagnik Chatterjee	Registered candidate contacted; no overnight final artifact found	E32-NO-RESULT

Entry / identity	Disposition / source status	Result IDs
E33 Japinder singh	Registered candidate contacted; no overnight final artifact found	E33-NO-RESULT
E34 1c1jp5dhwy	Score/project/reference located; complete final artifact or receipt not established	E34-HILL1
E35 aaronatlan	Score/project/reference located; complete final artifact or receipt not established	E35-HILL1
E36 advaith-appajodu	Score/project/reference located; complete final artifact or receipt not established	E36-HILL1
E37 alee792	Score/project/reference located; complete final artifact or receipt not established	E37-HILL1
E38 ashleychenyj	Score/project/reference located; complete final artifact or receipt not established	E38-HILL1
E39 claudiagaes	Score/project/reference located; complete final artifact or receipt not established	E39-HILL1
E40 claudioolmedo	Score/project/reference located; complete final artifact or receipt not established	E40-HILL1
E41 cloud-post-code	Score/project/reference located; complete final artifact or receipt not established	E41-HILL1
E42 connorbuchheit	Score/project/reference located; complete final artifact or receipt not established	E42-HILL1
E43 danamouk	Score/project/reference located; complete final artifact or receipt not established	E43-HILL1
E44 ebaenamar	Score/project/reference located; complete final artifact or receipt not established	E44-HILL1
E45 extraton618	Score/project/reference located; complete final artifact or receipt not established	E45-HILL1
E46 eychcue	Score/project/reference located; complete final artifact or receipt not established	E46-HILL1
E47 felixisaac	Reminder bounced; delivery failed; final packet missing	E47-NO-RESULT
E48 fkaita	Score/project/reference located; complete final artifact or receipt not established	E48-HILL1
E49 flaviabeppler	Score/project/reference located; complete final artifact or receipt not established	E49-HILL1
E50 frido22	Score/project/reference located; complete final artifact or receipt not established	E50-HILL1
E52 kevinsrun	Score/project/reference located; complete final artifact or receipt not established	E52-HILL1
E53 luizfelipebarbosa	Score/project/reference located; complete final artifact or receipt not established	E53-HILL1
E55 octavianboji	Score/project/reference located; complete final artifact or receipt not established	E55-HILL1
E56 ottogin	Organizer/reference account — not classified as independent entrant	E56-NO-RESULT
E58 sasap91	BB6 score; public repo exists but has no commits	E58-HILL1

Entry / identity	Disposition / source status	Result IDs
E59 sergeicu	Organizer/reference account — not classified as independent entrant	E59-NO-RESULT
E61 solal24	Score/project/reference located; complete final artifact or receipt not established	E61-HILL1, E61-HILL2
E63 vyahhi	BB6 held-out promising run; discovered repo predates event	E63-HILL1, E63-HILL2
E64 willow2014	Score/project/reference located; complete final artifact or receipt not established	E64-HILL1
E66 yavol	Score/project/reference located; complete final artifact or receipt not established	E66-HILL1, E66-HILL2, E66-HILL3
E67 Tom de Groot	Closed interest-only lead: no participation	E67-NO-RESULT
E68 Edward Hogan	Candidate/contact only; final packet location awaiting reply	E68-NO-RESULT
E69 Sri Kamarthapu	CLOSED interest — no work or teammate according to latest author reply	E69-NO-RESULT
E70 Arham Shuaib	Candidate/contact only; final packet location awaiting reply	E70-NO-RESULT
E71 Alexandros Kyriakakis	Interest only; no finished work reported	E71-NO-RESULT
E72 Martin Dvorak	Candidate/contact only; final packet location awaiting reply	E72-NO-RESULT
E73 Terry Wang — future competition interest	Missed this competition; interested in future challenges. No current entry.	E73-NO-RESULT
E74 Gopal Anantharaman — unmatched registration claim	September 18 says registered with another profile; no result packet received or account/team mapping verified	E74-NO-RESULT

OPDP scope and difficulty index

Each of the 70 records has a full problem profile, including the target even where the attempted result earns 0. Known-theorem rows have contextual D only; M2 ranking uses accepted families. P=published value retained; L=local single-assessor extension. Full factors, inputs, provenance and calculation are in OPDP-local-append.json and problems/<family>.txt.

Family	D / intrinsic	Origin	Contributors / rows
COLLATZ	772 / 6.8	P	E05, E08, E10, E66 / 4
KOBON	570 / 5.4	P	E08, E09, E10, E57, E37, E38, E39, E40, E48, E49, E52, E53, E55, E61, E64 / 88
RAMSEY-K4	840 / 8.4	L	E01, E07, E08, E12, E15, E43, E63, E66, E02 / 9
MATRIX-3	724 / 7.4	L	E04, E08, E10, E34, E36, E42, E02 / 10
GROTHENDIECK	561 / 5.4	P	E08, E10, E35, E44, E61, E02 / 6
BB6	974 / 10.0	L	E07, E08, E10, E13, E14, E41, E45, E46, E50, E58, E63, E66, E02 / 13
ERDOS3	674 / 7	P	E02, E10, E11, E65 / 1
DMS	574 / 5.4	P	E08 / 1
UNITARY	824 / 7.8	L	E03 / 1
HEILBRONN9	715 / 6.9	L	E12 / 1
CERNY	855 / 8.8	L	E17 / 1
M2-CONTAINERS	398 / 2.5	L	E06 / 1
M2-G-PM	317 / 1.5	L	E08 / 1
M2-E942	317 / 1.5	L	E08 / 1
M2-E44	317 / 1.5	L	E08 / 1
M2-E123	317 / 1.5	L	E08 / 1
M2-E918	317 / 1.5	L	E08 / 1
M2-E292	317 / 1.5	L	E08 / 1
M2-E395	317 / 1.5	L	E08 / 1
M2-E698	317 / 1.5	L	E08 / 1
M2-E939	317 / 1.5	L	E08 / 1
M2-E477	317 / 1.5	L	E08 / 1
M2-E295	272 / 1.0	L	E08 / 1
M2-E703	272 / 1.0	L	E08 / 1
M2-E748	272 / 1.0	L	E08 / 1
M2-E1136	272 / 1.0	L	E08 / 1
M2-E358	272 / 1.0	L	E08 / 1
M2-E619	272 / 1.0	L	E08 / 1
M2-E1148	272 / 1.0	L	E08 / 1
M2-OPTIONAL-E757	272 / 1.0	L	E08 / 1
M2-OPTIONAL-E261	272 / 1.0	L	E08 / 1
M2-OPTIONAL-E36	272 / 1.0	L	E08 / 1
M2-OPTIONAL-E649	272 / 1.0	L	E08 / 1
M2-OPTIONAL-E508	272 / 1.0	L	E08 / 1
M2-OPTIONAL-E1038	272 / 1.0	L	E08 / 1
M2-AP-CONCLUSION	272 / 1.0	L	E10, E65 / 2
M2-E3-THREE	424 / 2.8	L	E11, E65 / 2
M2-E3-DYADIC	436 / 3.1	L	E11, E65 / 2
M2-GOWERS4	424 / 2.8	L	E11 / 1
ERDOS169	538 / 4.9	P	E11, E02 / 2
CERNY-PART-1	425 / 3.0	L	E17 / 1
CERNY-PART-2	425 / 3.0	L	E17 / 1
CERNY-PART-3	425 / 3.0	L	E17 / 1

Family	D / intrinsic	Origin	Contributors / rows
CERNY-PART-4	425 / 3.0	L	E17 / 1
CERNY-PART-5	425 / 3.0	L	E17 / 1
CERNY-PART-6	425 / 3.0	L	E17 / 1
CERNY-PART-7	425 / 3.0	L	E17 / 1
CERNY-PART-8	425 / 3.0	L	E17 / 1
CERNY-PART-9	425 / 3.0	L	E17 / 1
OEIS-A100475	541 / 5.4	L	E02 / 1
OEIS-A060957	527 / 5.4	L	E02 / 1
OEIS-A000224	622 / 5.9	L	E02 / 1
LATIN-TABLEAU	650 / 6.1	L	E02 / 1
OPDP13-QUARTIC	644 / 5.0	L	E02 / 1
NO3-HALFTURN	330 / 3.8	L	E02 / 1
M2-BERNOULLI	346 / 1.9	L	E02 / 1
M2-QUATERNION	365 / 2.1	L	E02 / 1
M2-LCM-RECURRENCE	354 / 2.0	L	E02 / 1
M2-APERY	398 / 2.5	L	E02 / 1
M2-BASE3	346 / 1.9	L	E02 / 1
M2-THETA	365 / 2.1	L	E02 / 1
M2-WOLSTENHOLME	372 / 2.2	L	E02 / 1
M2-LCM-MATRIX	372 / 2.2	L	E02 / 1
M2-ERDOS887	372 / 2.2	L	E02 / 1
M2-POLYGON	354 / 2.0	L	E02 / 1
ERDOS1060	574 / 5.6	P	E02 / 1
ERDOS21-VARIANT	555 / 5.0	L	E02 / 1
ERDOS944	721 / 6.8	P	E02 / 1
RBM4-PARITY	727 / 5.9	L	E02 / 1
ERDOS829	530 / 5.4	P	E02 / 1

Sources, files and review boundaries

File summary coverage:353,195 /353,195;0 read errors. 347,231 complete small-file reads,5,818 bounded text prefixes,17 first-three-page PDF summaries and129 binary/role summaries. These are static summaries, not individually verified mathematical claims. Full original files remain in the local catalogue/archive library.

Primary sources used for the significant comparisons

Rules	Local EVENT/02-papers/competition_handbook_final.pdf, sections1,3–6.
Difficulty methodology	judging/Ulam_UnsolvedMath_ChatGPT_5.6_Sol_Ultra_Rationales_v1.0.json; frozen Atlas companion profiles and local append.
Ramsey prior	https://arxiv.org/pdf/2206.04036 — final McKay note, not just the headline bound.
Half-turn pre-event question	https://wwwhomes.uni-bielefeld.de/achim/no3in/symmetry_remarks.html — near-half-turn question before event.
Erdős–Lovász +3 question	https://arxiv.org/html/2606.24878v2 — explicit question after +4 lemma.
OEIS conjectures	https://oeis.org/A100475 and https://oeis.org/A060957 ; exact archived Formal Conjectures snapshots for A000224.
Uniform synthesis baseline	https://arxiv.org/abs/2403.13692 —22/48 coefficient; no proof of Matt21/48.
Erdős829 exact convention	Google DeepMind Formal Conjectures /ErdosProblems/829.lean, archived source correspondence.
Contestant evidence	Original archives, immutable commits, source messages, target correspondence, proof endpoints, supplied axiom/build receipts and independently written arithmetic/geometry checks. Per-file IDs in the226 entry notes.

Unresolved review dependencies

Missing Qichao original source ZIP; absent graphs for several hill-only accounts; no fresh full Lean builds; exact prior-formal-library novelty; eligibility and sealed independence; new difficulty median; Sakana late-source transfer decision. Chandra tensor truncation is resolved and all729identities and the symbolic family were independently checked. Every available contribution has a review disposition, numerical proposal and explicit remaining check.

Local tools

Desktop OpenMath Judging shortcut opens the native app. Person/entry review, Problem profiles and Proposed scores are the first tabs. Classified files shows each file’s summary and original content. Review notes save locally. Editable signoff tables are CSV; no email, LinkedIn message, submission, award, publication or access change was made.